

JATIN SATYAM

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EDUCATION

M.S. Robotics & Autonomous Systems (Systems Eng.), Arizona State University · GPA 3.67/4.0 Expected May 2027
B.E. Mechanical Engineering with Major (Honors) in Electric Vehicles, University of Mumbai 2020 – 2024

EXPERIENCE

Research Assistant Aug 2026 – Present

Robotics Systems Instructional Lab, Arizona State University

- Built the cell's ROS 2 Jazzy / Gazebo Harmonic model - both vendor arms and the conveyor merged into one 11-joint description - running the full wafer pick-transfer-place cycle autonomously in simulation, verified headless from six in-sim cameras
- Brought up a read-only real-to-sim digital shadow mirroring both physical arms live at 50 Hz - SCARA mapping verified to 0.1 degrees, tracking within 1.9 degrees / 7.6 mm on the bench; now building the same cell sequence in NVIDIA Isaac Sim with FANUC ER-4iA and LR Mate 200iD arms

Mechatronics Engineer Jan 2025 – Jul 2025

Paras Defence & Space Technologies Ltd., Navi Mumbai, India

- Designed and iterated Gifford-McMahon cryocooler mechanisms in SolidWorks and NX, refining moving-assembly tolerances and interfaces for reliable cold-head displacer motion and manufacturability
- Led performance testing of 3 cryogenic cooling assemblies across 25+ instrumented runs, characterizing thermal losses and translating findings into mechanism-level design changes that cut prototype rework 12%

CFD Engineer Aug 2024 – Nov 2024

MARS Exploration Pvt. Ltd, Remote

- Performed 3D CFD analysis of NACA 23012 and NACA 4412 airfoils in ANSYS Fluent 2024 R2 (10-degree AoA, 48 m/s) - geometry, boundary-layer-refined meshing with quality checks, solver setup, and convergence monitoring

R&D Engineer May 2023 – Aug 2023

TriQual Cryogenics Pvt Ltd, Navi Mumbai, India

- Contributed to mechanical design and vibration control for Stirling-cycle cryocoolers operating near 70 K, working alongside senior engineers on reciprocating-mechanism dynamics

PROJECTS

Aerial-Ground Cooperative Autonomy on Mars Terrain · ROS 2 Jazzy · Gazebo Harmonic · Python May 2026

- Built a 2-robot autonomy pipeline on real HiRISE Jezero Crater terrain in ROS 2 and Gazebo: a UAV surveys and builds a Bayesian log-odds occupancy grid (0.25 m/cell) that a 6-wheel rover navigates with A* replanning
- Implemented a vectorized terrain-relative navigation matcher - likelihood-field scoring of K=294 pose candidates at 2 Hz over 36 lidar rays - up to 24% lower localization error than odometry in feature-dense terrain, trailing odometry overall under sparse coverage (quantified in the report)

Autonomous Drone Search, Map & Precision Landing · ROS 2 Jazzy · PX4 · RTAB-Map March 2026

- Programmed a 9-state autonomous drone mission (PX4 SITL + ROS 2): search for rock formations, identify ArUco markers while orbiting, select the tallest cylinder, and land on top - 3/3 trials, graded 150/150 including extra credit
- Ran RTAB-Map RGB-D SLAM live during flight, producing a 43,283-vertex textured 3D mesh of the environment from 78 optimized poses

Natural-Language Robotic Manipulation with LLM Control · Python · OpenCV · LangChain December 2025

- Built an LLM-driven manipulation system on a physical Dobot Magician Lite: natural-language commands are parsed by Llama 3.3 70B on Groq, objects located via HSV computer vision, and pick-and-place executed through validated motion primitives
- Calibrated camera-to-robot coordinates with a least-squares affine transformation, achieving +/-2 mm positioning precision in the xy-plane at 15-30 s per pick-and-place cycle

SKILLS

Programming: Python (NumPy, SciPy, OpenCV, Open3D), C++, C, MATLAB, Bash, SQL
Software: ROS 2 (Humble/Jazzy), Gazebo, NVIDIA Isaac Sim, PX4 SITL, MuJoCo, Simulink, RTAB-Map, Docker, FANUC (ER-4iA, LR Mate 200iD), SolidWorks, Siemens NX, ANSYS, Git & GitHub
Other Skills: Kinematics, Motion Planning, A* Planning, SLAM, Occupancy Mapping, PID, LQR, State-Space Modeling

AWARDS & RECOGNITION

- 2nd Prize, National Level Project Poster Competition, IEI-FCRIT 2024 - gyro motor rotor (CALIBRE 2k24)
- Winner, MLH Best Use of Auth0 - Meridian, Innovation Hacks 2.0, ASU, Apr 2026
- Co-inventor, invention disclosure filed with ASU's Skysong Innovations - SipSafe Smart Cup, 2026